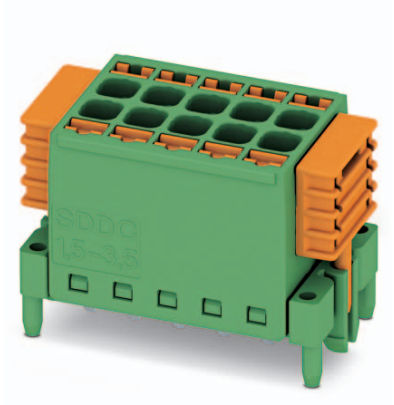
2016-11-09
Product version 00
Document revision 0

# Data sheet

**Order No.: 1848710**

**Type: SDDC 1,5/ 9-PV-3,5**

**Plug component, Push-in spring connection**




The illustration shows the 5-pos. version

## 1 Main features




- Number of positions 9
- Conductor cross section 1.5 mm²
- Color green
- Pitch 3.50 mm
- Connection method Push-in spring connection
- Nominal current 8 A
- Nominal voltage 160 V
- Connection direction 0 °
- Type of packaging packed in cardboard

## 2 Your advantages

- SKEDD direct plug-in technology enables flexible positioning on the PCB
- Reduced component and process costs: simple insertion by hand and vibration-resistant connection
- Time saving push-in connection, tools not required
- Intuitive use through colour coded actuation lever
- Quick and convenient testing using integrated test option

Make sure you always use the latest documentation.
It can be downloaded at: phoenixcontact.net/product/1848710

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# 4 3D model in PDF can be activated (Acrobat Reader only)



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# 5 Item properties

| | |
|---|---|
| Order No. | 1848710 |
| Type | SDDC 1,5/ 9-PV-3,5 |
| Range of articles | SDDC 1,5/..-PV |
| Pitch | 3.50 mm |
| Number of positions | 9 |
| Connection method | Push-in spring connection |
| Mounting type | SKEDD - Direct plug-in technology |
| Pin layout | Linear double pinning |

## 5.1 Material data

| Material of metal parts | | |
|---|---|---|
| Note | WEEE/RoHS-compliant, whisker-free acc. to IEC 60068-2-82/JEDEC JESD 201 | |
| Contact material | Cu alloy | |
| Terminal point surface | Sn 4 µm ... 8 µm | |
| Surface contact area | Sn 4 µm ... 8 µm | |
| Surface characteristics | hot-dip tin-plated | |

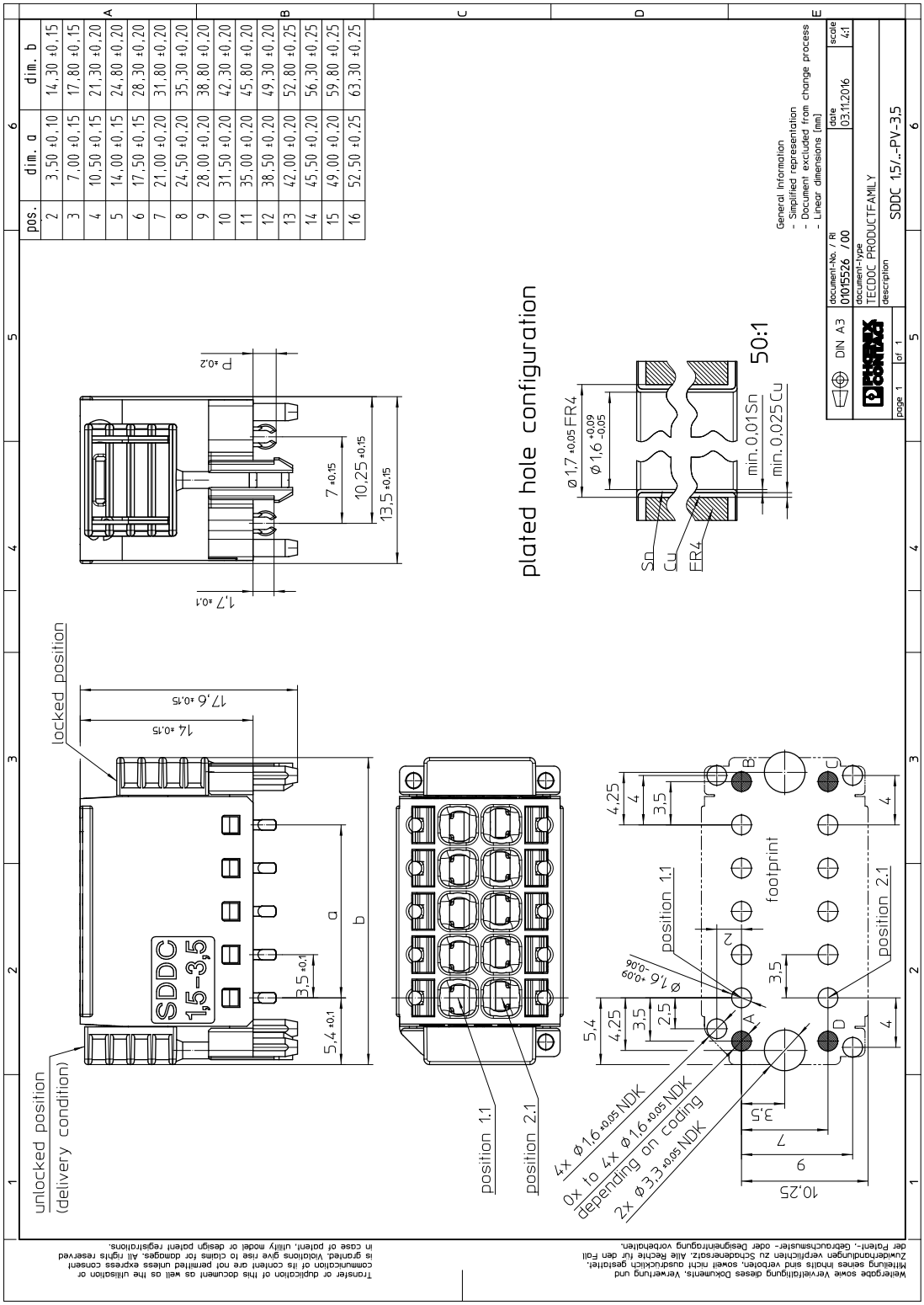
| Insulating material data | Housing | |
|---|---|---|
| Insulating material | PA | PBT |
| CTI according to IEC 60112 | 600 | 600 |
| Flammability rating according to UL 94 | V0 | V0 |
| Color | green (6021) | |
| Glow wire flammability index GWFI according to EN 60695-2-12 | 850 | |
| Glow wire ignition temperature GWIT according to EN 60695-2-13 | 775 | |
| Temperature for the ball pressure test according to EN 60695-10-2 | 125 °C | |

## 5.2 Dimensions

| | |
|---|---|
| Dimension a | 28 mm |
| Length | 13.5 mm |
| Width | 38.8 mm |
| Total height | 17.6 mm |
| Pin dimensions | 0,9 x 2,0 mm |
| Pin spacing | 7.00 mm |
| Hole diameter | 1.6 mm |

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# 6 Series drawing




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# 7 Packaging information

| | |
|---|---|
| Type of packaging | packed in cardboard |
| Pieces per package | 100 |

# 8 Application

## 8.1 Temperature limit values

| | |
|---|---|
| Ambient temperature (storage/transport) | -40 °C ... 70 °C |
| Ambient temperature (assembly) | -5 °C ... 100 °C |
| Ambient temperature (operation) | -40 °C |

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# 9 Mechanical tests

## 9.1 Pull-out test

| Termination and connection method: pull-out test | |
|---|---|
| Specification | IEC 60999-1:1999-11 |
| Result | Test passed |
| Conductor cross section/conductor type/tractive force actual value | 0.2 $mm^2$ / solid / > 10 N |
| Conductor cross section/conductor type/tractive force actual value | 0.2 $mm^2$ / stranded / > 10 N |
| Conductor cross section/conductor type/tractive force actual value | 1.5 $mm^2$ / solid / > 40 N |
| Conductor cross section/conductor type/tractive force actual value | 1.5 $mm^2$ / stranded / > 40 N |
| Conductor cross section/conductor type/tractive force actual value | AWG 16 / stranded / > 40 N |

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# 10 Electrical tests

## 10.1 Electrical data

| | |
|---|---|
| Rated current / conductor cross section | 8 A / 1.5 mm² |
| Rated insulation voltage (III/2) | 160 V |
| Rated surge voltage (III/2) | 2.5 kV |
| Contact resistance | 1.4 mΩ |
| Degree of pollution | 2 |

## 10.2 Air and creepage distances

| | | | |
|---|---|---|---|
| Component | Plug component | | |
| Specification | IEC 60664-1:2007-04 | | |
| Mains type | unearthed mains | | |
| Insulating material group | I | | |
| Comparative tracking index (IEC 60112:2003-01) | CTI 600 | | |
| Rated insulation voltage | 160 V | 160 V | 400 V |
| Rated surge voltage | 2.5 kV | 2.5 kV | 2.5 kV |
| Degree of pollution | 3 | 2 | 2 |
| Overvoltage category | III | III | II |
| Minimum clearance case A (inhomogeneous field) | 1.5 mm | 1.5 mm | 1.5 mm |
| Minimum value of the creepage path requirement in acc. with table | 2 mm | 0.8 mm | 2 mm |

## 10.3 Electrical function

| | |
|---|---|
| Specification | IEC 60999-1:1999-11 |
| Result | Test passed |
| Voltage drop | Voltage drop (U) after the load ≤ 15 mV |
| Conductor cross section, flexible | |
| Conductor cross section, solid | |

## 10.4 Temperature cycles

| | |
|---|---|
| Specification | IEC 60999-1:1999-11 |
| Result | Test passed |
| Voltage drop | Voltage drop (U) after the load ≤ 22.5 mV or 1.5 x $U_{\text{after 24 h}}$<br>The small value is to be used. |
| Test current | 4 A DC |
| Temperature cycles | |
| Conductor cross section, flexible | |
| Conductor cross section, solid | |

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# 11 Current carrying capacity/derating curves

| | |
|---|---|
| Specification | IEC 61984:2008-10 |
| Note | Representation based on IEC 60512-5-2:2002-02 |
| Reduction factor | 0.8 |
| Number of positions | See diagram |
| Conductor cross section | 1.5 mm² |

**Type: SDDC 1,5/...-PV-3,5**

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# 12 Environmental and durability tests

## 12.1 Vibration test

| | |
|---|---|
| Specification | IEC 60068-2-6:2007-12 |
| Result | Test passed |
| Frequency | 10 - 150 - 10 Hz |
| Sweep speed | 1 octave/min |
| Amplitude | 0.35 mm (10 - 60.1 Hz) |
| Acceleration | 5 g (60.1 - 150 Hz) |
| Test duration per axis | 2.5 h |
| Test directions | X-, Y- and Z-axis |
| Requirements | |
| Note | |
| | |

# 13 Classification for connectors

| | |
|---|---|
| Specification | IEC 61984:2008-10 |
| Main features | Connectors without switching capacity (COC) |
| Construction form | Fixed connectors |
| Strain relief elements | without strain relief |
| Connection method | Can be reconnected |
| Protection against electric shock | Not encapsulated - touch-proof when inserted |
| Protective conductor | without PE |
| Lock | no |
| Connection method | Screwless terminal points |

# 14 Approvals

# 15 Commercial data

| | |
|---|---|
| Order No. | 1848710 |
| Type | SDDC 1,5/ 9-PV-3,5 |
| Pieces per package | 100 |
| Net weight | 2.22 g |
| GTIN | 4055626307183 |
| Customs tariff number | Information that applies locally, see link on page 1 |
| Country of origin | Information that applies locally, see link on page 1 |

# 16 Project documentation

| | |
|---|---|
| Project documentation | M6 |

# 17 Accessories

| Description | Order No. | Type |
|---|---|---|
| Coding profile, is inserted into the hole in the plug, red insulating material | 1985564 | CP-PT 1,5 |
| | 0804073 | SK 3,5/2,8:FORTL.ZAHLEN |
| | 0825121 | SK 2,8 REEL P3,5 WH CUS |
| | 0803883 | SK U/2,8 WH:UNBEDRUCKT |

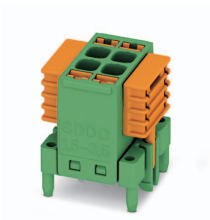
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## 1848710 SDDC 1,5/ 9-PV-3,5

| Description | Order No. | Type |
|---|---|---|
| | 0805205 | SK 2,8 WH:REEL |
| | 1944372 | MPS-MT 1-S |
| | 1982800 | MPS-MT 1-S4-B RD |
| Crimping pliers, for ferrules without insulating collar according to DIN 46228 Part 1 and ferrules with insulating collar according to DIN 46228 Part 4, 0.25 mm² ... 6.0 mm², lateral entry, trapezoidal crimp | 1212034 | CRIMPFOX 6 |
| | 3203037 | AI 0,25- 8 YE |
| | 3200014 | AI 0,5 - 8 WH |
| | 3200881 | AI 0,5 - 8 WH -1000 |
| | 3201275 | AI 0,5 -10 WH |
| | 3201288 | AI 0,75-10 GY |
| | 3200182 | AI 1 -10 RD |
| | 3202481 | A 0,5 - 8 |
| | 3202494 | A 0,5 -10 |
| | 3202504 | A 0,75- 8 |
| | 3200234 | A 0,75-10 |
| | 3202517 | A 1 - 8 |
| | 3200250 | A 1 -10 |

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## 17.1 Combination tests



### SDDC 1,5/..-PV

| | | | | |
|---|---|---|---|---|
| Specification | IEC 61984 | | | |
| **Mechanical tests (A)** | | | | |
| Insertion/withdrawal force per position | approx. 8 N / 6 N | | | |
| Polarization when inserted<br>Requirement > 20 N | Test passed | | | |
| Contact holder in insert<br>Requirements > 20 N | | | | |
| **Endurance tests (B)** | | | | |
| Contact resistance $R_1$ | 1.4 mΩ | | | |
| Insertion/withdrawal cycles | 25 | | | |
| Contact resistance $R_2$ | 1.5 mΩ | | | |
| Rated impulse voltage at sea level<br>Voltage waveform ≥ (1.2/50 µs) | 2.95 kV | | | |
| Power-frequency withstand voltage<br>Voltage waveform ≥ (50/60 Hz) | 1.39 kV | | | |
| Insulation resistance<br>Requirements > 5 MΩ | > 0.2 TΩ | | | |
| **Thermal tests (C)** | | | | |
| Tested number of positions | 16 | | | |
| Tested conductor cross section | 1.5 $mm^2$ | | | |
| Test current | 8 A | | | |
| Upper limiting temperature<br>Requirements < 100°C | Test passed | | | |
| **Climatic tests (D)** | | | | |
| Test sequence 1: low temperature storage | -40 °C/2 h | | | |
| Test sequence 2: heat storage | 100 °C/168 h | | | |
| Test sequence 3: noxious gas storage (ISO 6988) | 0.2 $dm^3$ $SO_2$ on 300 $dm^3$/ 40 °C/1 cycle | | | |
| Rated impulse voltage at sea level<br>Voltage waveform ≥ (1.2/50 µs) | 2.95 kV | | | |
| Power-frequency withstand voltage<br>Voltage waveform ≥ (50/60 Hz) | 1.39 kV | | | |
| **Environmental and endurance tests (E)** | | | | |
| Specification | IEC 60529:1989-11 + AMD 1:1999-11 + AMD 2:2013-08 | | | |
| Type of protection (when plugged in) | Finger safety with IP20 test finger | | | |

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